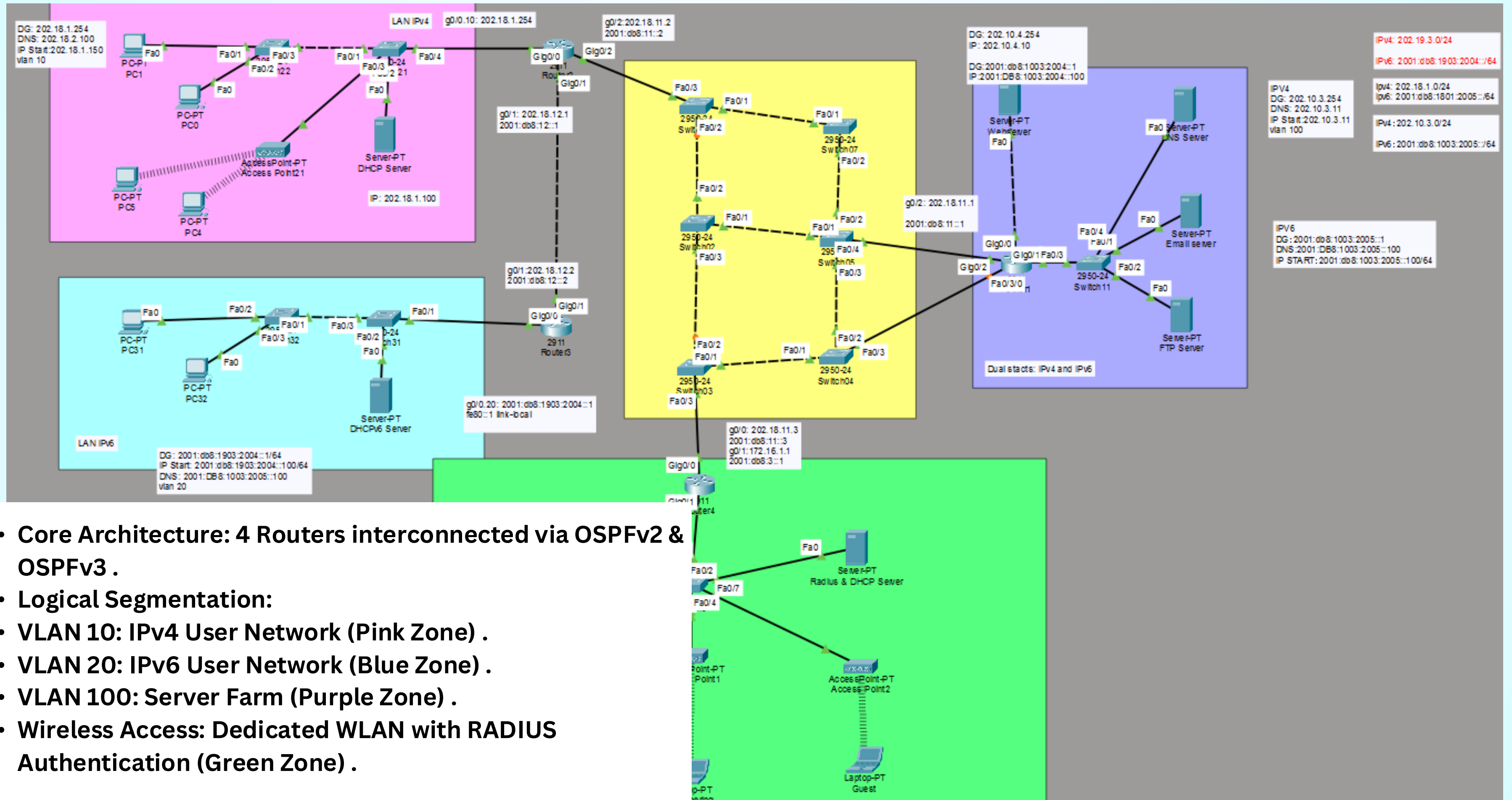


Dual Stack IPv4 and IPv6 High Availability Network Design and Simulation Using Cisco Packet Tracer

Group 35



Network Design Overview



Addressing Plan

Device	Interface	IPv4 Address/Prefix	IPv6 Address/Prefix	Default Gateway
Router 1 (Purple)	G0/0	202.10.4.254/24	2001:DB8:1003:2004::1/64	N/A
	G0/1.100	202.10.3.254/24	2001:DB8:1003:2005::1/64	N/A
	G0/2	202.18.11.1/24	2001:DB8:11::1/64	N/A
Router 2 (Pink)	G0/0.10	202.18.1.254/24	N/A	N/A
	G0/1	202.18.12.1/24	2001:DB8:12::1/64	N/A
	G0/2	202.18.11.2/24	2001:DB8:11::2/64	N/A
Router 3 (Blue)	G0/0.20	N/A	2001:DB8:1903:2004::1/64	N/A
	G0/1	202.18.12.2/24	2001:DB8:12::2/64	N/A
Router 4 (Green)	G0/0	202.18.11.3/24	2001:DB8:11::3/64	N/A
	G0/1	172.16.1.1/24	2001:DB8:3::1/64	N/A

Addressing Plan

PC0	NIC	DHCP Assign	N/A	202.18.1.254
PC1	NIC	DHCP Assign	N/A	202.18.1.254
PC4	NIC	DHCP Assign	N/A	202.18.1.254
PC5	NIC	DHCP Assign	N/A	202.18.1.254
DHCP Server	NIC	202.18.1.100/24	N/A	202.18.1.254
PC31	NIC	N/A	SLAAC from 2001:DB8:1903:2004::/64	2001:DB8:1903:2004::1
PC32	NIC	N/A	SLAAC from 2001:DB8:1903:2004::/64	2001:DB8:1903:2004::1
DHCPv6	NIC	N/A	2001:DB8:1903:2004::10/64	2001:DB8:1903:2004::1
Web Server	NIC	202.10.4.10/24	2001:DB8:1003:2004::100/64	IPv4: 202.10.4.254 IPv6: 2001:DB8:1003:2004::1
DNS Server	NIC	202.10.3.11/24	2001:DB8:1003:2005::100/64	IPv4: 202.10.3.254 IPv6: 2001:DB8:1003:2005::1
Email Server	NIC	202.10.3.12/24	2001:DB8:1003:2005::101/64	IPv4: 202.10.3.254 IPv6: 2001:DB8:1003:2005::1
FTP Server	NIC	202.10.3.13/24	2001:DB8:1003:2005::102/64	IPv4: 202.10.3.254 IPv6: 2001:DB8:1003:2005::1
Radius & DHCP Server	NIC	172.16.1.10 /24	SLAAC from 2001:DB8:3::/64	IPv4: 172.16.1.1 IPv6: 2001:DB8:3::1
Laptop Marketing	Wireless	DHCP Assign	SLAAC from 2001:DB8:3::/64	IPv4: 172.16.1.1 IPv6: 2001:DB8:3::1
Laptop Engineering	Wireless	DHCP Assign	SLAAC from 2001:DB8:3::/64	IPv4: 172.16.1.1 IPv6: 2001:DB8:3::1
Laptop Guest	Wireless	DHCP Assign	SLAAC from 2001:DB8:3::/64	IPv4: 172.16.1.1 IPv6: 2001:DB8:3::1

Additional Segmentation

VLAN

Switch Configuration

```
Switch>enable
Switch#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#vlan 10
Switch(config-vlan)#name VLAN0010
Switch(config-vlan)#exit
Switch(config)#interface fastEthernet0/1
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 10
Switch(config-if)#exit
Switch(config)#interface fastEthernet0/2
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 10
Switch(config-if)#exit
```

VLAN 10 (IPv4 User LAN):

- Created to isolate standard IPv4 user traffic.

```
Switch>enable
Switch#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#
Switch(config)#vlan 20
Switch(config-vlan)# name IPv6_User_LAN
Switch(config-vlan)# exit
Switch(config)#
Switch(config)#interface fastEthernet0/2
Switch(config-if)# switchport mode access
Switch(config-if)# switchport access vlan 20
Switch(config-if)# exit
Switch(config)#
Switch(config)#interface fastEthernet0/1
Switch(config-if)# switchport mode trunk
Switch(config-if)# exit
Switch(config)#
Switch(config)#interface fastEthernet0/3
Switch(config-if)# switchport mode trunk
Switch(config-if)# exit
Switch(config)#
```

VLAN 20 (IPv6 User LAN):

- Created specifically for IPv6-native clients.

VLAN

Switch Configuration

```
Switch>enable
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#
Switch(config)#vlan 100
Switch(config-vlan)#name VLAN0100
Switch(config-vlan)#exit
Switch(config)#
Switch(config)#interface fastEthernet0/1
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 100
Switch(config-if)#exit
Switch(config)#
Switch(config)#interface fastEthernet0/2
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 100
Switch(config-if)#exit
Switch(config)#
Switch(config)#interface fastEthernet0/4
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 100
Switch(config-if)#exit
```

VLAN 100 (Server Farm):

- Isolates critical servers (DNS, Web, FTP) from user traffic to improve security.

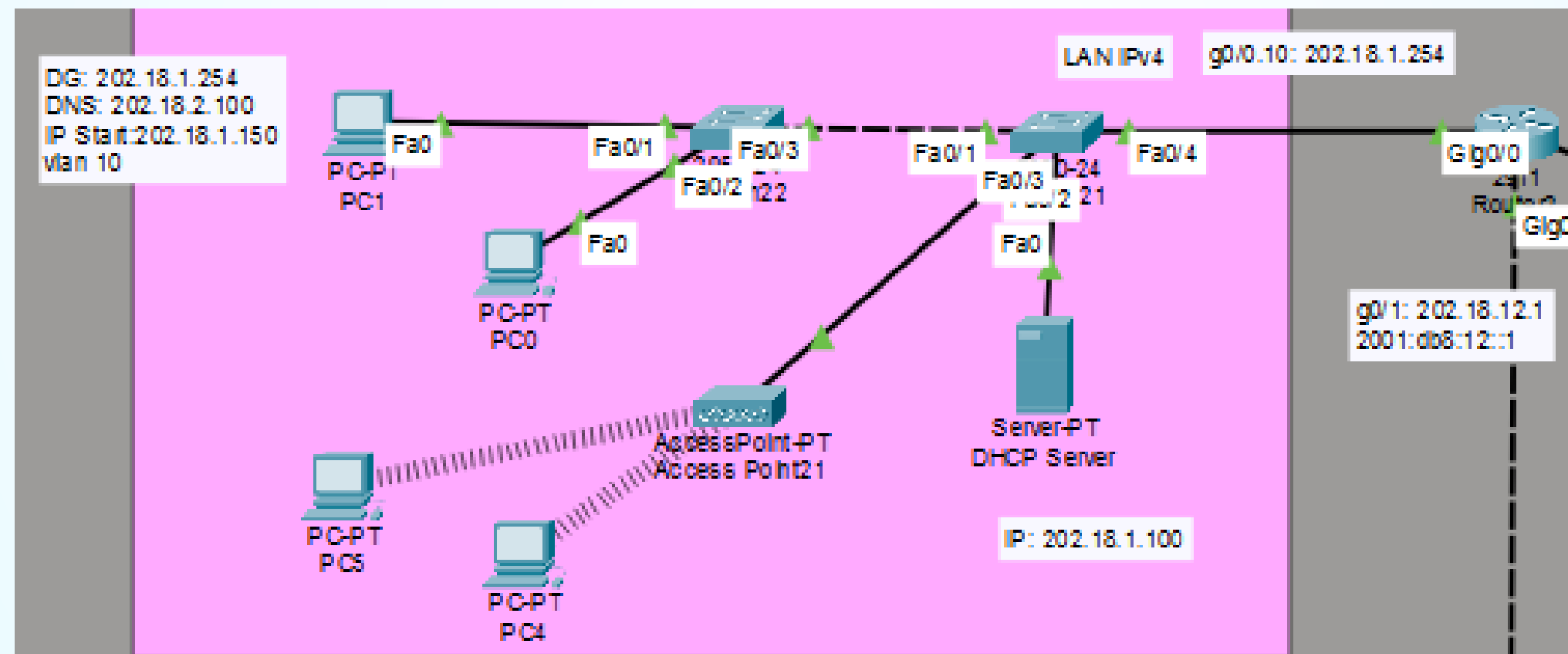
```
Switch>enable
Switch#show interfaces trunk
Port      Mode      Encapsulation  Status        Native vlan
Fa0/1     desirable n-802.1q       trunking      1
Fa0/2     on        802.1q         trunking      1
Fa0/3     on        802.1q         trunking      1

Port      Vlans allowed on trunk
Fa0/1     1-1005
Fa0/2     1-1005
Fa0/3     1-1005
```

Trunk Configuration:

- Switch ports configured with IEEE 802.1Q encapsulation.
- Verified to carry traffic for VLANs 10, 20, and 100 simultaneously.

User LAN IPv4 (VLAN 10)



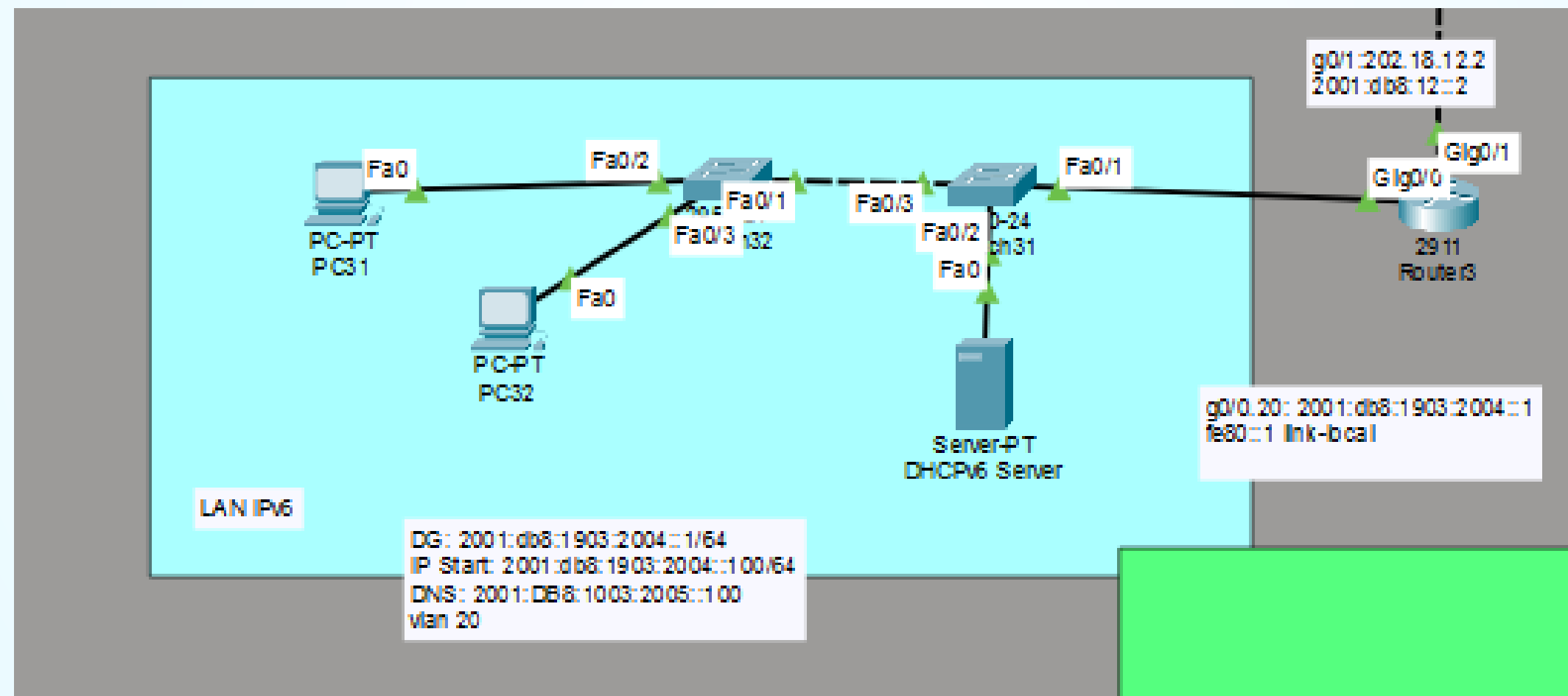
Routing Configuration

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#
Router(config)#interface GigabitEthernet0/0
Router(config-if)# no shutdown
Router(config-if)#exit
Router(config)#
Router(config)#interface GigabitEthernet0/0.10
Router(config-subif)# encapsulation dot1Q 10
Router(config-subif)# ip address 202.18.1.254 255.255.255.0
Router(config-subif)# ip ospf 1 area 0
Router(config-subif)# no shutdown
Router(config-subif)#exit
Router(config)#interface GigabitEthernet0/2
Router(config-if)# ip address 202.18.11.2 255.255.255.0
Router(config-if)# ipv6 address 2001:DB8:11::2/64
Router(config-if)# ip ospf 1 area 0
Router(config-if)# ipv6 ospf 1 area 0
Router(config-if)# no shutdown
Router(config-if)#exit
Router(config)#router ospf 1
Router(config-router)# router-id 2.2.2.2
Router(config-router)# network 202.18.1.0 0.0.0.255 area 0
Router(config-router)# network 202.18.11.0 0.0.0.255 area 0
Router(config-router)#exit
Router(config)#
Router(config)#ipv6 router ospf 1
Router(config-rtr)# router-id 2.2.2.2
Reload or use "clear ipv6 ospf process" command, for this to take
effect
Router(config-rtr)#exit
Router(config)#
```

Router 2: Core Backbone Hub

- **Role:** Central Hub for OSPF Area 0.
- **Stack:** Dual-Stack (IPv4/IPv6) to support communication between all departments.
- **Identity:** Stable Router ID (2.2.2.2).
- **Status:** Full connectivity with R1, R3, and R4.

User LAN IPv6 (VLAN 20)



Routing Configuration

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#
Router(config)#ipv6 unicast-routing
Router(config)#
Router(config)#interface GigabitEthernet0/0.20
Router(config-subif)# encapsulation dot1Q 20
Router(config-subif)# ipv6 address 2001:DB8:1903:2004::1/64
Router(config-subif)# ipv6 enable
Router(config-subif)# ipv6 ospf 1 area 0
Router(config-subif)# exit
Router(config)#
Router(config)#interface GigabitEthernet0/1
Router(config-if)# ipv6 address 2001:DB8:12::2/64
Router(config-if)# ipv6 enable
Router(config-if)# ipv6 ospf 1 area 0
Router(config-if)# exit
Router(config)#
Router(config)#ipv6 router ospf 1
Router(config-rtr)# router-id 3.3.3.3
Reload or use "clear ipv6 ospf process" command, for this to take effect
Router(config-rtr)# exit
Router(config)#
Router(config)#end
```

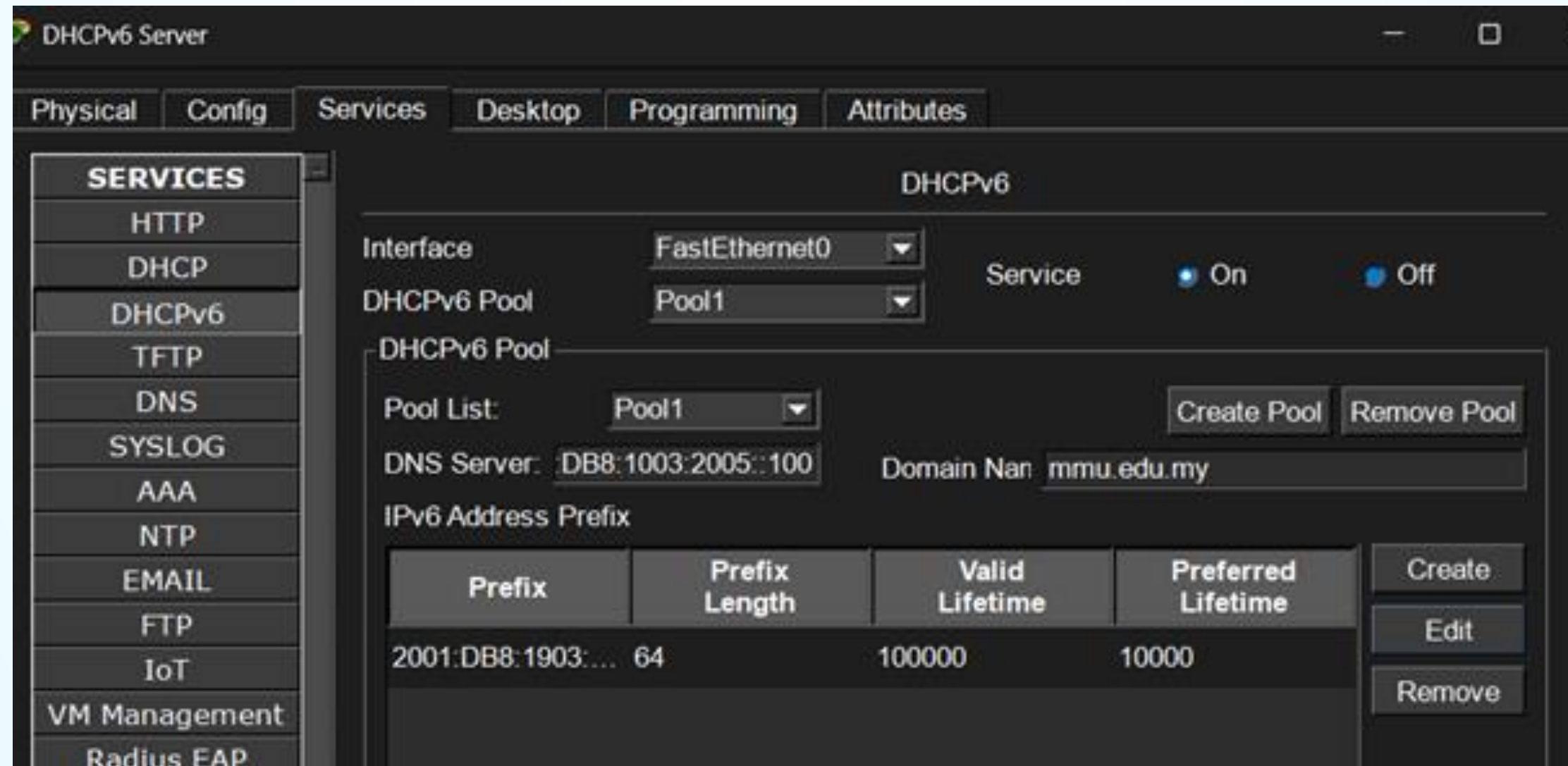
Router 3: User Gateway

- **Role:** Connects the Blue area to the core.
- **Stack:** Dual-Stack support for cross-network communication.
- **Identity:** Stable Router ID (3.3.3.3).
- **Status:** FULL adjacency with R2 via OSPFv2/v3.

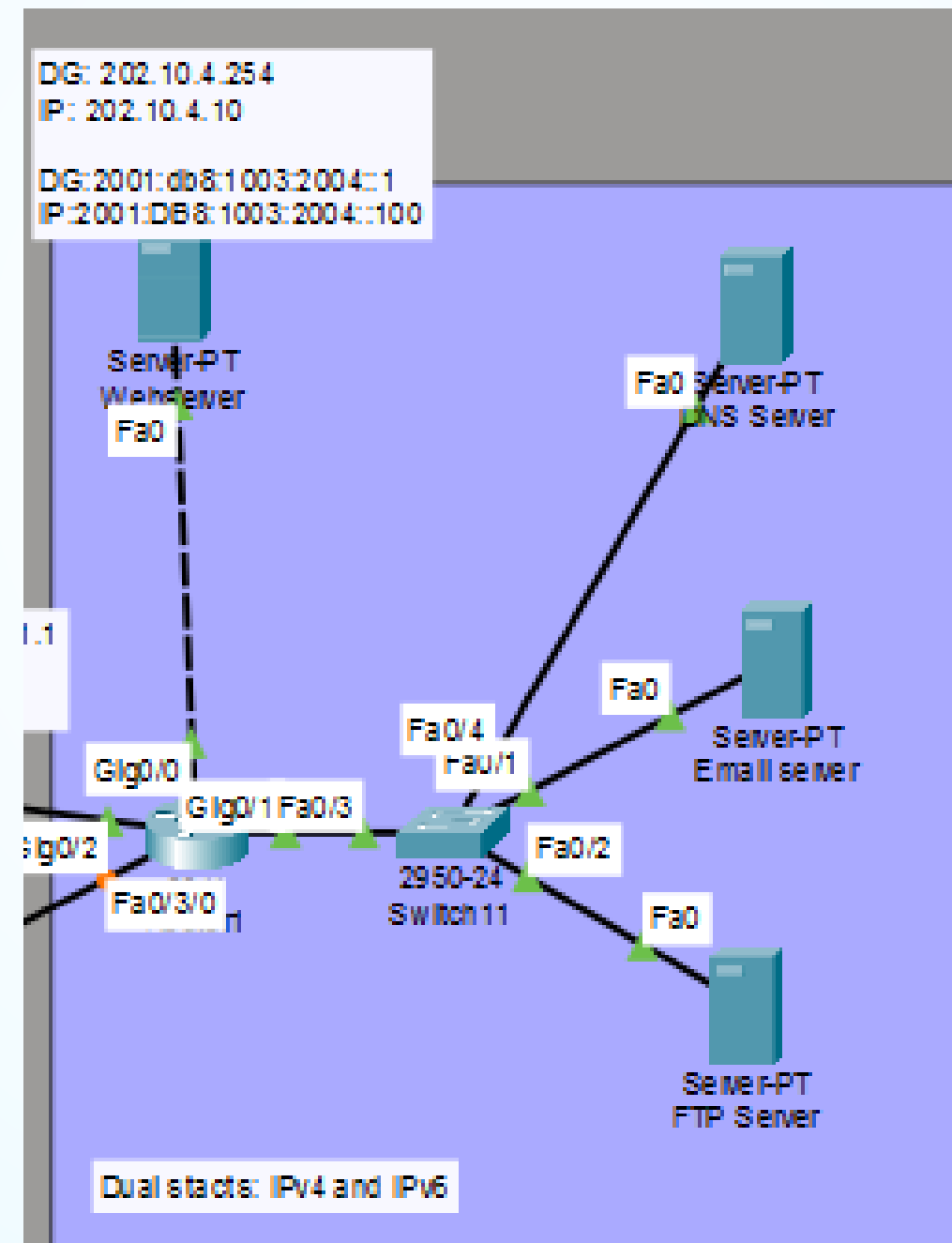
DHCPv6

Configuration

A Stateful DHCPv6 approach was implemented.
This ensures consistent IP address assignment.



Dual Static: IPv4 & IPv6 (VLAN 100)



Routing Configuration

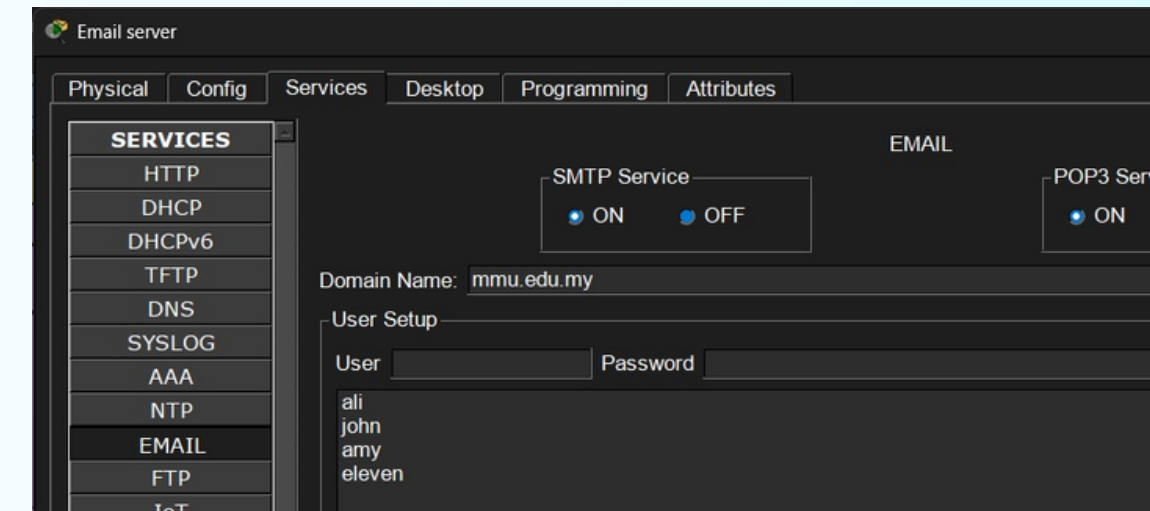
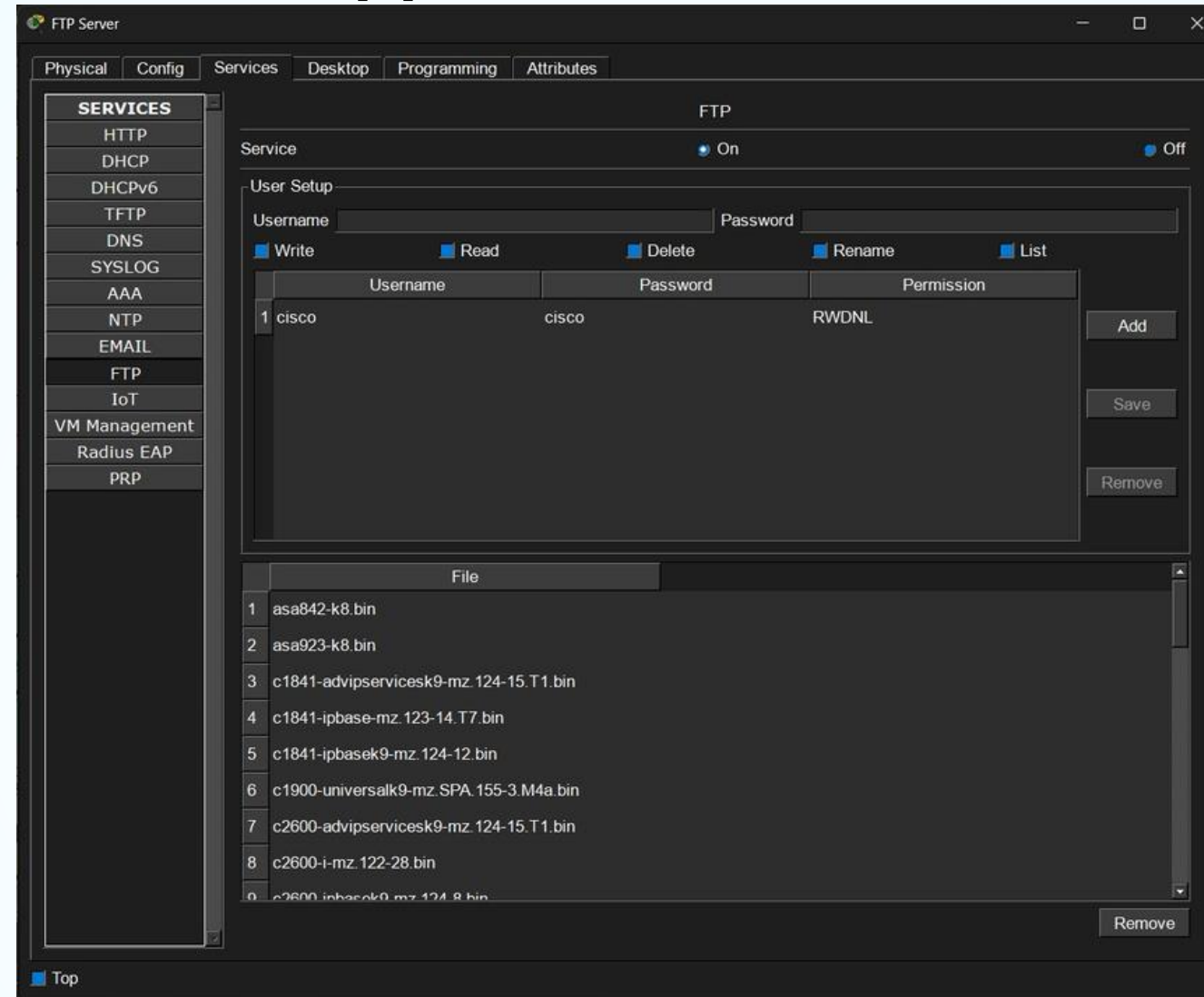
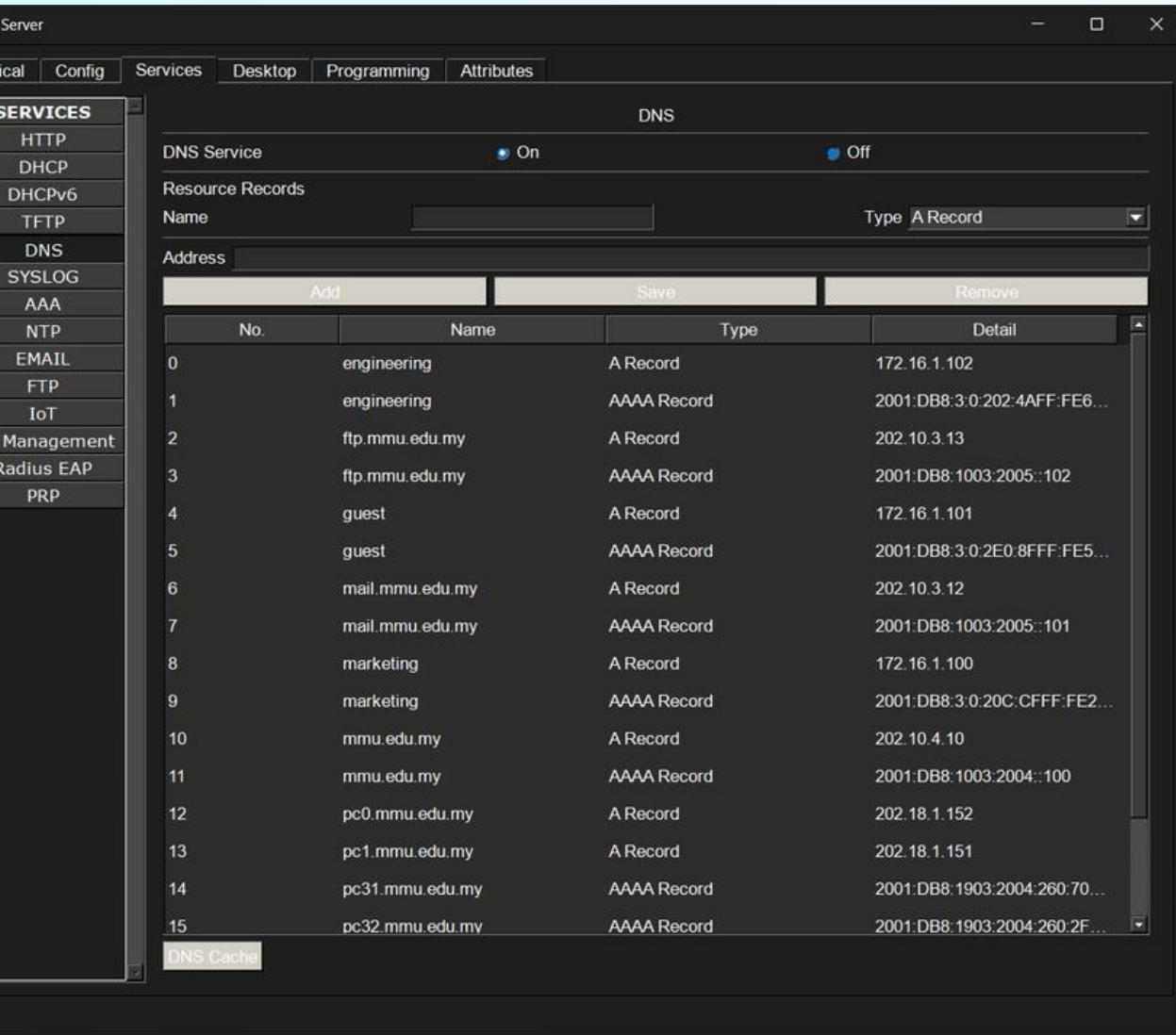
```
Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#ipv6 unicast-routing
Router(config)#interface GigabitEthernet0/1
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#interface GigabitEthernet0/1.100
Router(config-subif)#encapsulation dot1Q 100
Router(config-subif)#ip address 202.10.3.254 255.255.255.0
Router(config-subif)#ipv6 address 2001:DB8:1003:2005::1/64
Router(config-subif)#ipv6 ospf 1 area 0
Router(config-subif)#no shutdown
Router(config-subif)#exit
Router(config)#interface GigabitEthernet0/0
Router(config-if)#ip address 202.10.4.254 255.255.255.0
Router(config-if)#ipv6 address 2001:DB8:1003:2004::1/64
Router(config-if)#ip ospf 1 area 0
Router(config-if)#ipv6 ospf 1 area 0
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#interface GigabitEthernet0/2
Router(config-if)#ip address 202.10.11.1 255.255.255.0
Router(config-if)#ipv6 address 2001:DB8:11::1/64
Router(config-if)#ip ospf 1 area 0
Router(config-if)#ipv6 ospf 1 area 0
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#router ospf 1
Router(config-router)#router-id 1.1.1.1
Router(config-router)#network 202.10.3.0 0.0.0.255 area 0
Router(config-router)#network 202.10.4.0 0.0.0.255 area 0
Router(config-router)#network 202.10.11.0 0.0.0.255 area 0
Router(config-router)#exit
Router(config)#
Router(config)#ipv6 router ospf 1
Router(config-rtr)#router-id 1.1.1.1
Reload or use "clear ipv6 ospf process" command, for this to take effect
Router(config-rtr)#exit
Router(config)#
```

Router 1: Server Department Gateway

- **Role:** Connects the Server Department to the core.
- **Stack:** Dual-Stack support for simultaneous IPv4/IPv6 communication.
- **Routing:** Forwards departmental traffic to R2 to reach the Servers.
- **Identity:** Stable Router ID (1.1.1.1)
- **Status:** FULL adjacency with R2 and R4 via OSPFv2/v3.

Server Configuration

Static Application Services



DNS Service (202.10.3.11):

- Acts as the network "phonebook," mapping domain names to IP addresses.

Records Created:

- mmu.edu.my → Web Server (202.10.4.10)
- ftp.mmu.edu.my → FTP Server (202.10.3.13)
- mail.mmu.edu.my → Email Server (202.10.3.12)

Impact:

- Ensures users can access all services using user-friendly names instead of memorizing IP addresses .

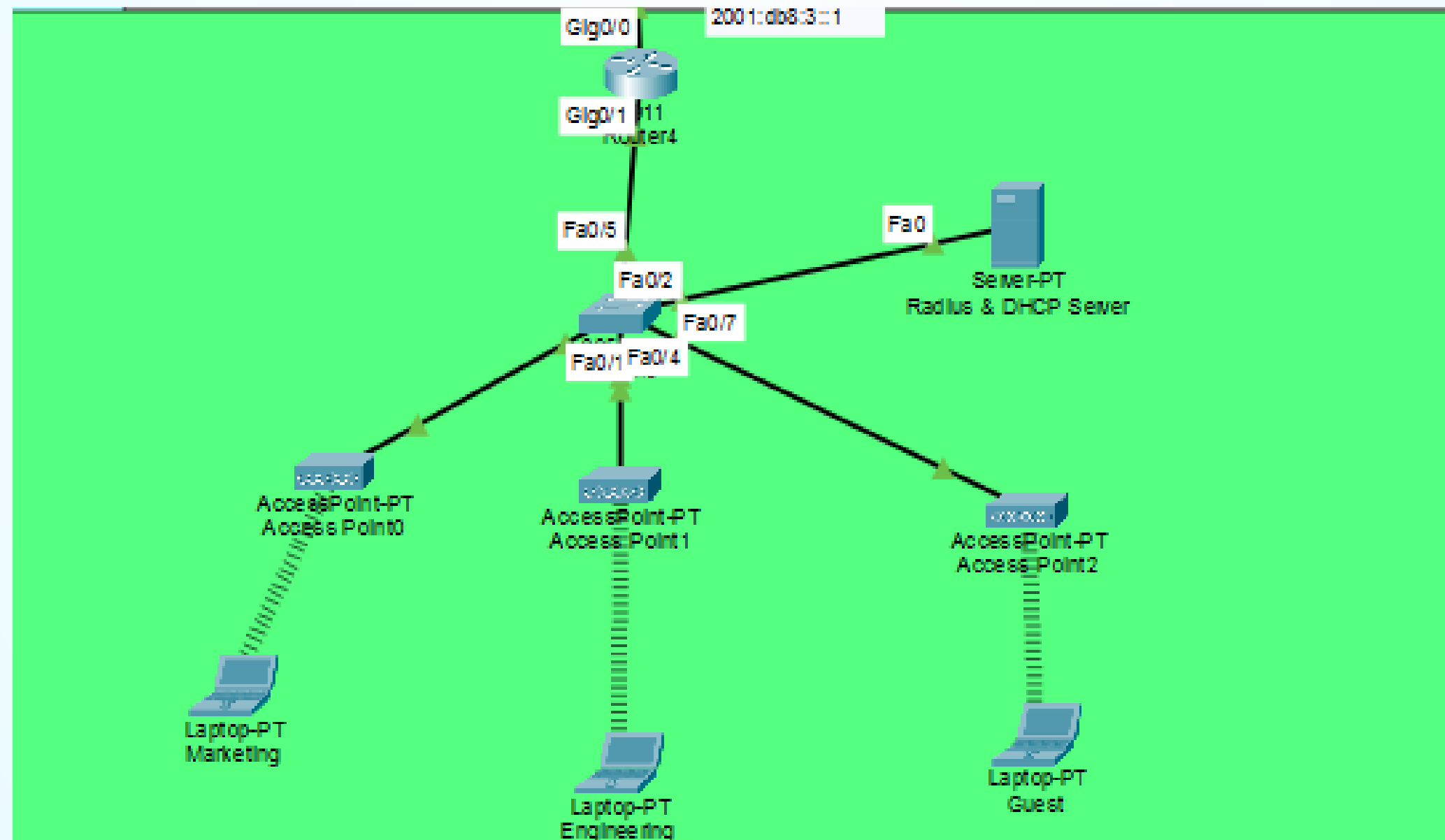
FTP Server (202.10.3.13):

- Function: Enables secure file transfer for staff and students.
- Security: Configured with User Authentication (User: cisco) to prevent unauthorized access.
- Permissions: User granted full read/write access (RWDNL).

Email Server (202.10.3.14):

- Function: Enables send and receive emails.

Wireless & Radius Security



Routing Configuration

```
User Access Verification

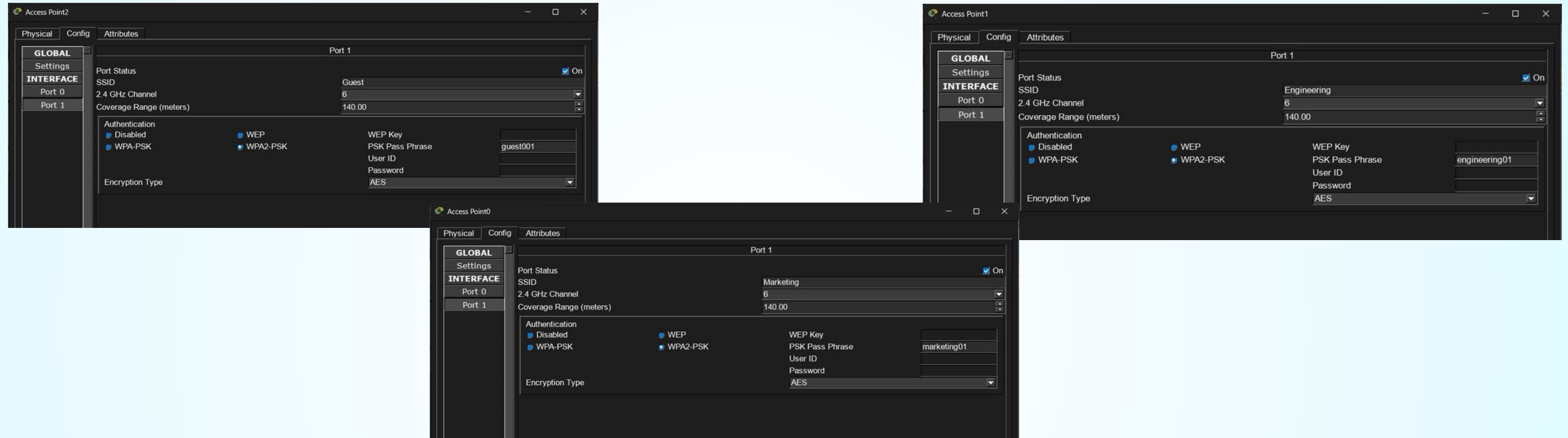
Username: ml
Password:
R4>enable
Username:
Password:
R4#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R4(config)#interface GigabitEthernet0/0
R4(config-if)# ip address 202.18.11.3 255.255.255.0
R4(config-if)# ipv6 address 2001:DB8:11::3/64
R4(config-if)#ip ospf 1 area 0
R4(config-if)# ipv6 ospf 1 area 0
R4(config-if)#no shutdown
R4(config-if)#exit
R4(config)#interface GigabitEthernet0/1
R4(config-if)# ip address 172.16.1.1 255.255.255.0
R4(config-if)# ipv6 address 2001:DB8:3::1/64
R4(config-if)# no shutdown
R4(config-if)#exit
R4(config)#router ospf 1
R4(config-router)# router-id 4.4.4.4
R4(config-router)# network 202.18.11.0 0.0.0.255 area 0
R4(config-router)#exit
R4(config)#
R4(config)#ipv6 router ospf 1
R4(config-rtr)# router-id 4.4.4.4
Reload or use "clear ipv6 ospf process" command, for this to take
effect
R4(config-rtr)#exit
```

Router 4: Wireless & Security Gateway

- **Role:** Connects the Wireless Department (AP & Laptop) to the core.
- **Stack:** Dual-Stack support for wireless IPv4/IPv6 clients.
- **Identity:** Stable Router ID (4.4.4.4).
- **Status:** FULL adjacency with R2 and via OSPFv2/v3.

Server Configuration

Static Application Services



Departmental Isolation:

- Created 3 separate WLANs: Marketing, Engineering, and Guest.
- Prevents unauthorized access between departments.

Security Configuration:

- WPA2-PSK: Configured unique passphrases for each group (e.g., marketing01, guest001).

Wireless & RADIUS Security

RADIUS Server Configuration

The screenshot shows the 'RADIUS & DHCP Server' configuration window with the 'Services' tab selected. The 'AAA' service is enabled. The 'Network Configuration' section shows a table with one entry: Client Name '4', Client IP '172.16.1.1', Server Type 'Radius', and Key 'cisco'. The 'User Setup' section shows a table with three entries: Username 'u1', Password '2345'; Username 'guest', Password 'gt'; and Username 'm1', Password '1234'.

Client Name	Client IP	Server Type	Key
1 4	172.16.1.1	Radius	cisco

Username	Password
1 u1	2345
2 guest	gt
3 m1	1234

RADIUS Authentication:

- Used to authenticate wireless users and store credentials centrally.
- Router 4 is registered as a trusted RADIUS client.

Wireless Security:

- Ensures only authorized users (e.g., 'u1', 'guest') can access the network.

Wireless LAN

DHCP Server Configuration

The screenshot shows the 'Services' tab in the 'Radius & DHCP Server' configuration window. The 'DHCP' section is active, showing configuration for the 'FastEthernet0' interface. The 'Service' is set to 'On'. The 'Pool Name' is 'serverPool'. The 'Default Gateway' is '172.16.1.1'. The 'DNS Server' is '202.16.3.11'. The 'Start IP Address' is '172.16.1.1' and the 'Subnet Mask' is '255.255.255.0'. The 'Maximum Number of Users' is '156'. The 'TFTP Server' and 'WLC Address' are both '0.0.0.0'. Below the configuration fields is a table with columns for 'Pool Name', 'Default Gateway', 'DNS Server', 'Start IP Address', 'Subnet Mask', 'Max User', 'TFTP Server', and 'WLC Address'. The table contains one entry for 'serverPool'.

Pool Name	Default Gateway	DNS Server	Start IP Address	Subnet Mask	Max User	TFTP Server	WLC Address
serverPool	172.16.1.1	202.16.3.11	172.16.1.1	255.255.255.0	156	0.0.0.0	0.0.0.0

DHCP Server:

- Allocates IPv4 addresses from the 172.16.1.1/24 pool.
- Automatically provides Gateway and DNS settings to end users.

Console, VTY, and enable-secret security Configuration

```
R4#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R4(config)#aaa new-model
R4(config)#aaa authentication login TELNET_AUTH group radius local
R4(config)#aaa authentication login WLAN_AUTH group radius local
R4(config)#radius-server host 172.16.1.10 key cisco
R4(config)#line vty 0 4
R4(config-line)#login authentication TELNET_AUTH
R4(config-line)#transport input telnet
R4(config)#enable secret class123
R4(config)#service password-encryption
```

Testing

Testing

The screenshot displays the 'Wireless Network Monitor v1.0' interface. At the top, there are tabs for 'Link Information', 'Connect', and 'Profiles'. Below these are buttons for 'Back', 'Statistics', and 'Save to Profile'. The main section is titled 'Wireless Network Status' and contains a table with the following data:

Wireless Network Status			
Radio Band	20MHz	Network Type	Wireless-G
Wireless Network Name	Marketing	IP Address	172.16.1.102
Wireless Mode	Infrastructure	Subnet Mask	255.255.255.0
Wide Channel	N/A	Default Gateway	172.16.1.1
Standard Channel	1 - 2.412GHz	DNS1	202.10.3.11
Security	WPA2-Personal	MAC Address	0080.5C4A.85B4
Authentication	Auto		

Below the table, there are two signal strength indicators: 'Signal Strength' and 'Link Quality', both showing full bars. To the right, it says 'Adapter is Active'. At the bottom, it identifies the device as a 'Wireless-N Notebook Adapter' with model number 'WPC300N'.

Wireless Security:

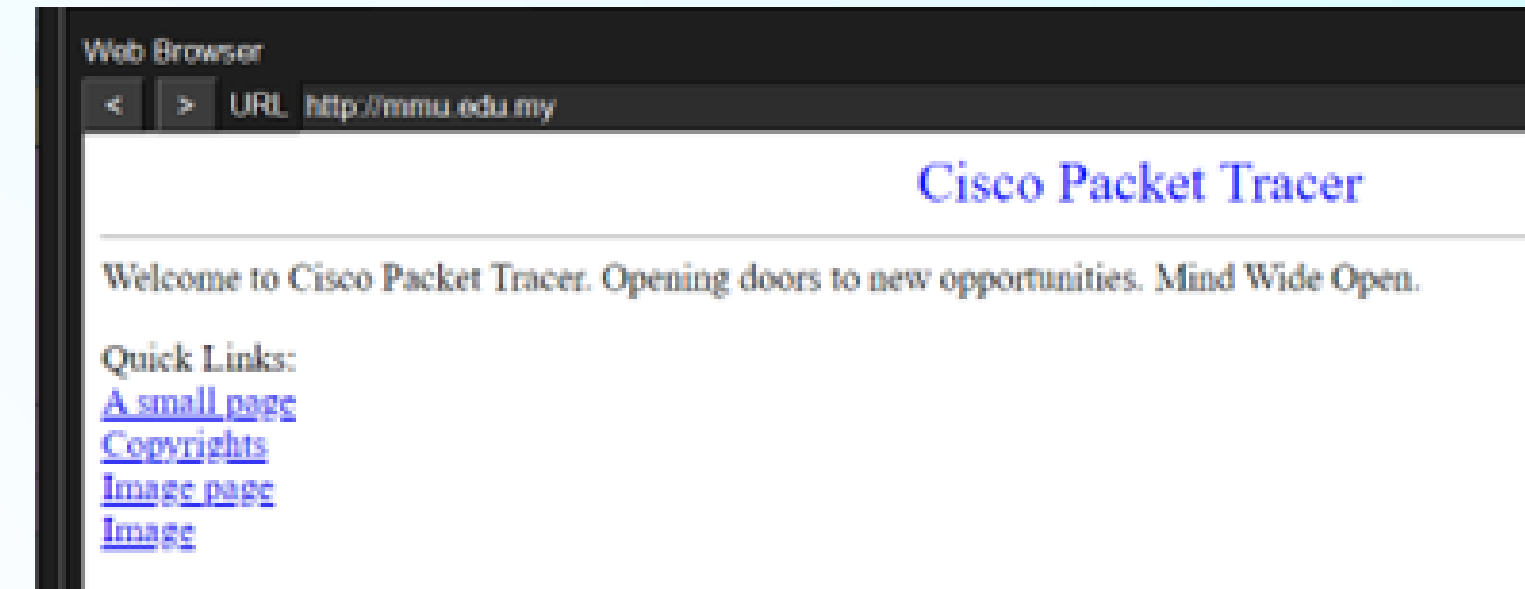
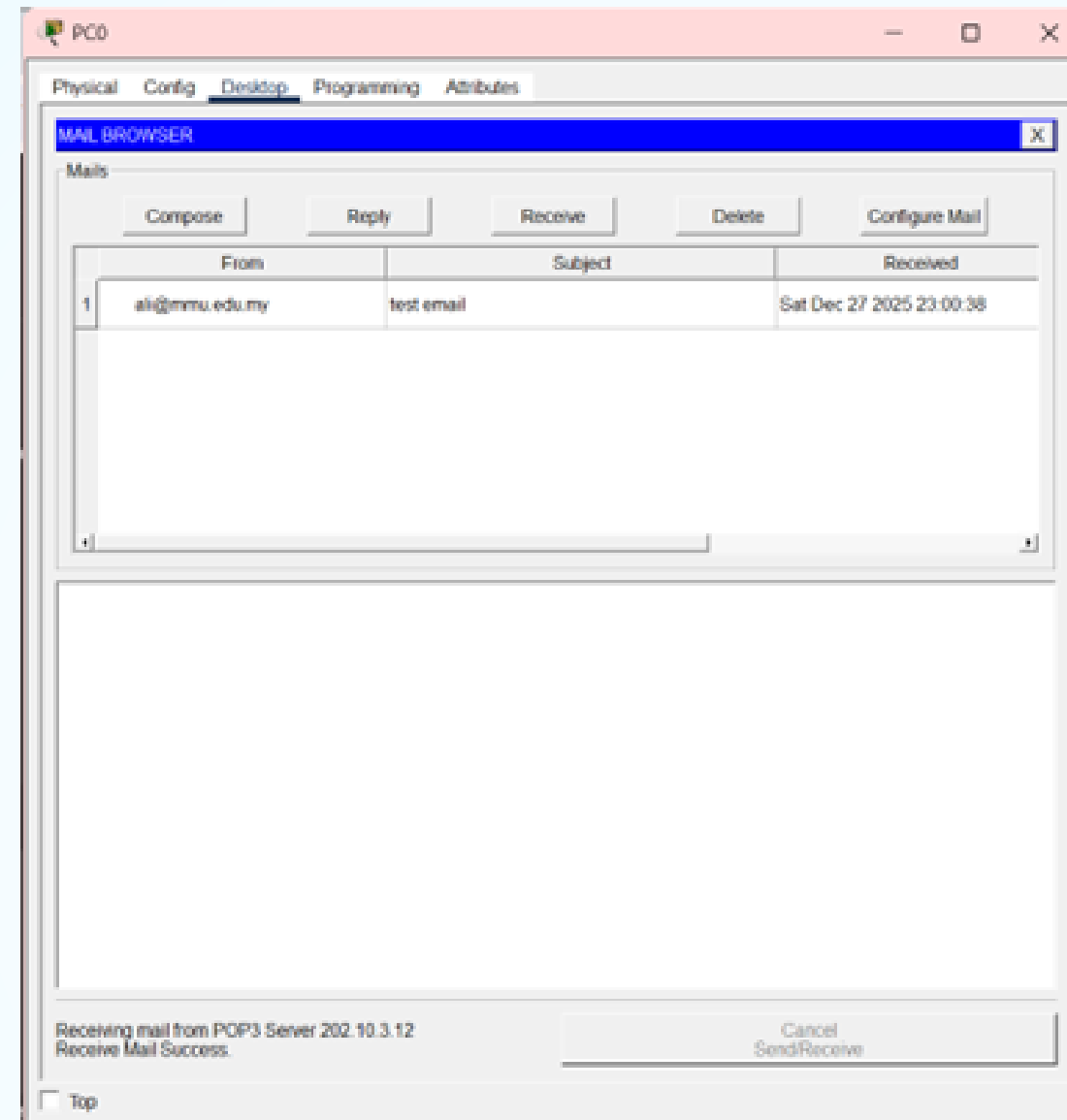
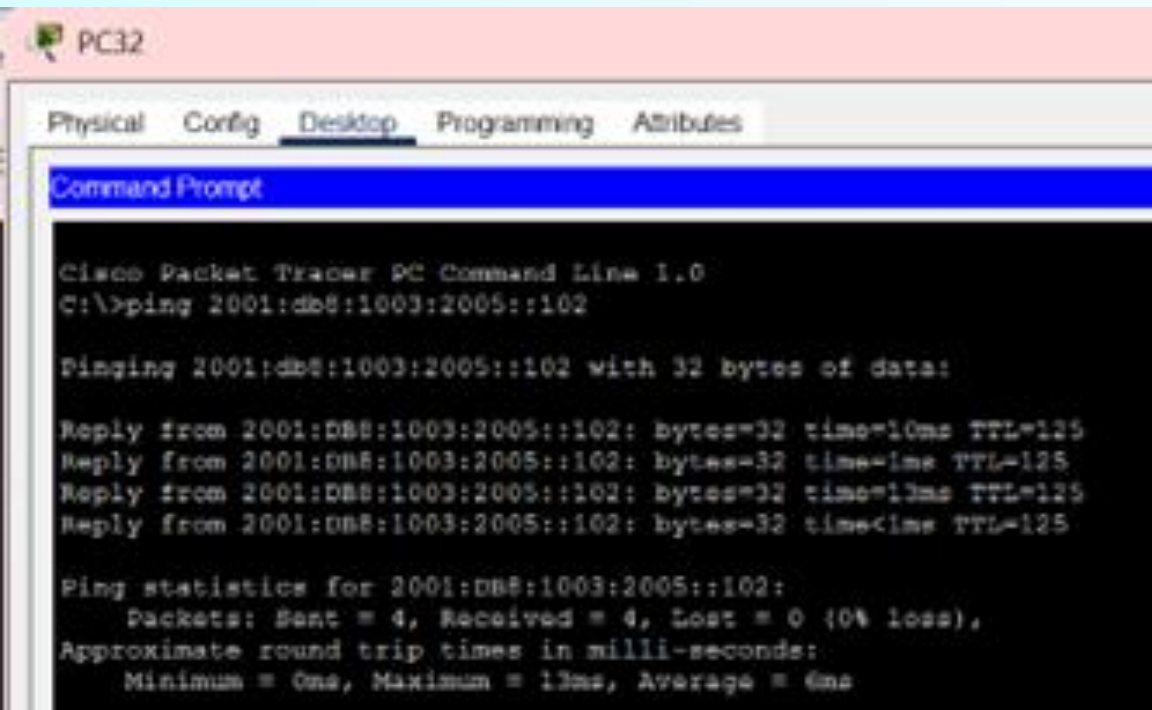
- Successful wireless connectivity was verified by the Laptop receiving a dynamic IP via DHCP. Furthermore, RADIUS authentication was confirmed by a successful Telnet login to Router 4 using credentials stored on the central AAA server

```
Cisco Packet Tracer PC Command Line 1.0
C:\>telnet 172.16.1.1
Trying 172.16.1.1 ...Open

User Access Verification

Username: m1
Password:
R4>
```

Testing



Connectivity (Ping):

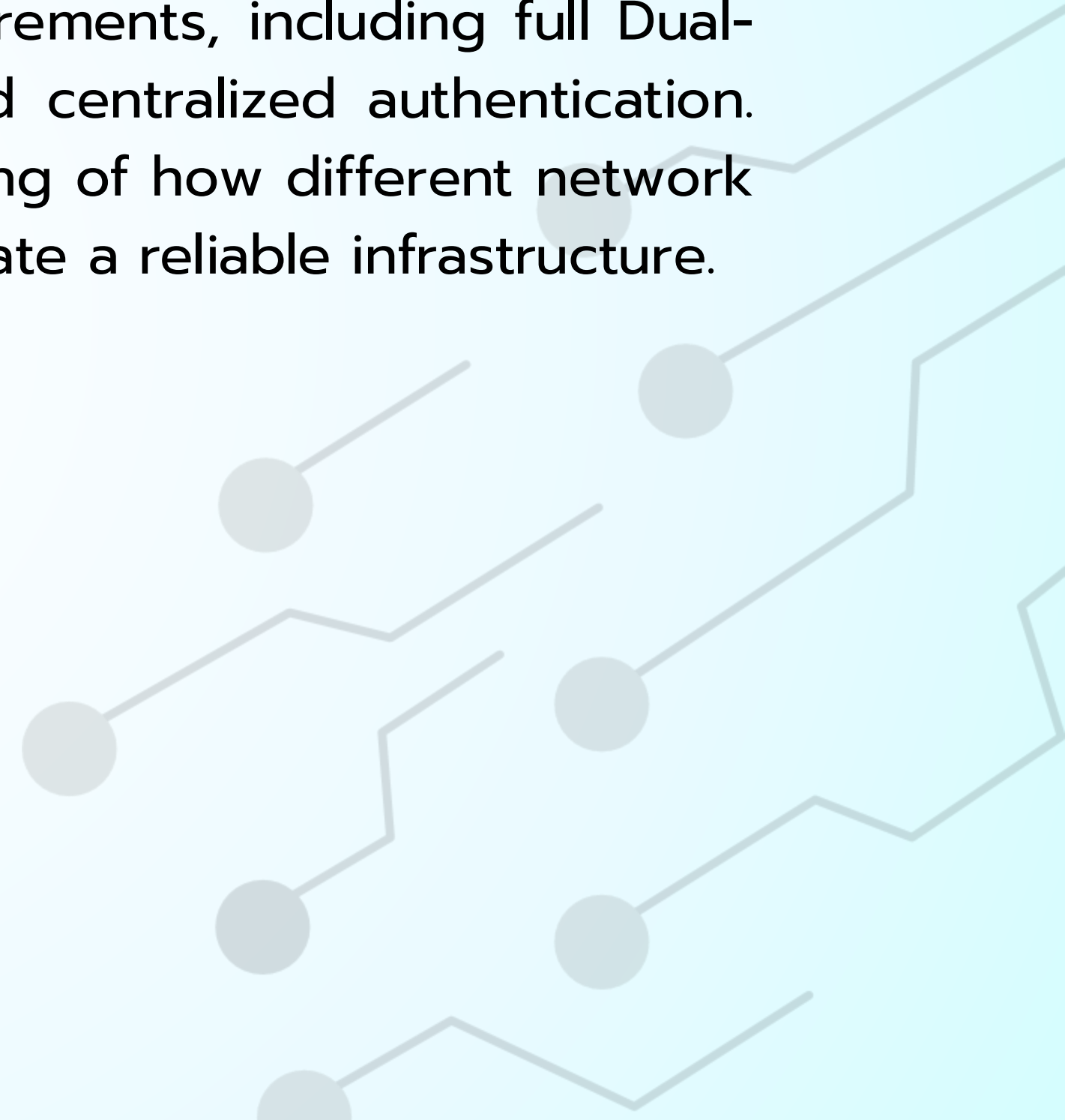
- 0% Packet Loss verified between User PC and Core Servers .
- Confirms OSPF routing and VLANs are working.

Service Verification:

- Web: Browser successfully loaded the departmental site.
- Email: Successful Send & Receive test between users ali@mmu.edu.my and bob.

Conclusion

To conclude, this project allowed us to simulate a realistic enterprise network environment. We successfully met all the design requirements, including full Dual-Stack support, secure segmentation using VLANs, and centralized authentication. This experience significantly improved our understanding of how different network protocols—like OSPF and RADIUS—work together to create a reliable infrastructure.





Thank you

